Axioms, Conditioning & Bayes' Theorem: 20 Worked Problems

Kolmogorov Framework & Event Spaces · Bayesian Updating & Conditioning · Classical Bayesian Traps

This chapter is where most candidates lose points — not from unfamiliar formulas, but from trusting intuition over the axioms. Every problem below carries a plain-language *Intuition* analogy (because Bayes' theorem should feel like "updating a hunch," not symbol-pushing) alongside a full derivation and, where relevant, a *Speed trick*. Work the three sections in order: the traps in Section III only make sense once Sections I–II are automatic.

I. Kolmogorov Framework & Event Spaces

I.A — Axioms & Sample Spaces

Problem 1 (Verifying Kolmogorov's axioms). A junior analyst proposes two probability assignments on the biased die $\Omega = \{1, \ldots, 6\}$:
(a) $P(1..6) = 0.10, 0.15, 0.20, 0.25, 0.20, 0.15$.
(b) $P(1..6) = 0.30, 0.25, -0.05, 0.20, 0.15, 0.15$.
For each, state which Kolmogorov axiom (if any) is violated.

Solution. Kolmogorov's axioms: (1) $P(A) \geq 0$ for every event A; (2) $P(\Omega) = 1$; (3) countable additivity for disjoint events. (a) Every individual value is ≥ 0 (axiom 1 holds), but $\sum P(i) = 0.10 + 0.15 + 0.20 + 0.25 + 0.20 + 0.15 = 1.05 \neq 1$. **Axiom 2 fails.** Fix: divide every value by 1.05 to renormalize. (b) $\sum P(i) = 0.30 + 0.25 - 0.05 + 0.20 + 0.15 + 0.15 = 1.00$ (axiom 2 holds!), but $P(3) = -0.05 < 0$. **Axiom 1 fails.**

★ *Intuition:* A probability assignment is like a pie chart: axiom 1 says no slice can have negative area, axiom 2 says the slices must exactly cover the whole pie (not 105% of it, not 95%). Case (a) drew a pie that's slightly too big; case (b) drew one slice with "negative area," which is nonsense even though the total area happens to work out.

► *Speed trick / key lesson:* Always sum first (checks axiom 2 in one line) and scan for negatives second (checks axiom 1). Real data errors overwhelmingly produce (a)-type failures from rounding or double-counted frequencies.

Problem 2 (Additivity requires disjointness — deriving a bound from it). Events A_1, A_2 (not assumed disjoint) satisfy $P(A_1) = 0.6$, $P(A_2) = 0.5$. Prove A_1, A_2 cannot be disjoint, and find the tightest lower bound on $P(A_1 \cap A_2)$ that the axioms alone force.

Solution. If A_1, A_2 were disjoint, axiom 3 would give $P(A_1 \cup A_2) = P(A_1) + P(A_2) = 1.1$. But axiom 2 plus monotonicity forces $P(A_1 \cup A_2) \leq P(\Omega) = 1$. Since $1.1 > 1$, disjointness is impossible. By inclusion–exclusion, $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2) \leq 1$, so

$$P(A_1 \cap A_2) \;\geq\; P(A_1) + P(A_2) - 1 = 0.6 + 0.5 - 1 = \boxed{0.1}.$$

★ *Intuition:* Think of two overlapping spending categories that each eat "60%" and "50%" of a fixed 100%-budget pie: they *must* overlap by at least 10 percentage points, or the pie would need to be 110% — the axioms are just accounting rules that refuse to let probability exceed 1.

► *Speed trick / key lesson:* Whenever two probabilities sum to more than 1, you instantly know a nontrivial lower bound on their intersection — no extra information needed. This is the "reverse" of Boole's inequality (Problem 4) and is itself sometimes called the Bonferroni lower bound for two events.

Problem 3 (Event spaces / σ-algebras generated by a partition). Let Ω be the outcomes of 3 coin flips, so $|\Omega| = 8$. (a) How many events exist in the full power-set σ-algebra? (b) A trader only ever reasons about $X = \#\text{heads} \in \{0, 1, 2, 3\}$. How many events live in the (coarser) σ-algebra generated by this partition of Ω into 4 blocks?

Solution. (a) The power set of an 8-element set has $2^8 = \boxed{256}$ events. (b) A σ-algebra generated by a partition into k blocks contains exactly the unions of subsets of those blocks (including the empty union and the full union), so it has 2^k events. Here $k = 4$: $2^4 = \boxed{16}$ events.

★ *Intuition:* Full Ω is like being able to ask about *every possible* combination of coin sequences (256 distinct yes/no questions you could pose). Restricting to "#heads only" is like wearing glasses that blur individual sequences together into just 4 buckets $\{0, 1, 2, 3\}$ — you've thrown away 240 of the 256 questions because you can no longer distinguish, say, HTT from THT (both "1 head").

► *Speed trick / key lesson:* "#events in a σ-algebra generated by a k-block partition $= 2^k$" is a one-line fact that's easy to forget under pressure — don't reach for $k!$ or $\binom{2^n}{k}$; it's always 2^k.

I.B — Union Bounds: Boole's & Bonferroni Inequalities

Problem 4 (Bonferroni's two-sided sandwich). Four events $A_1, \ldots, A_4$ each have $P(A_i) = 0.2$, and every pairwise intersection has $P(A_i \cap A_j) = 0.03$ (no independence assumed). Using only Bonferroni's first two orders, sandwich $P(\bigcup A_i)$ between an upper and lower bound.

Solution. Let $S_1 = \sum P(A_i) = 4(0.2) = 0.8$ and $S_2 = \sum_{i<j} P(A_i \cap A_j) = \binom{4}{2}(0.03) = 6(0.03) = 0.18$. **Upper bound (Boole, truncate after S_1):** $P(\bigcup A_i) \leq S_1 = 0.8$. **Lower bound (truncate after $S_1 - S_2$):** $P(\bigcup A_i) \geq S_1 - S_2 = 0.8 - 0.18 = \boxed{0.62}$. So $0.62 \leq P(\bigcup A_i) \leq 0.80$.

★ *Intuition:* Imagine estimating total unique website visitors by adding up per-page visit counts (S_1) — you're overcounting anyone who viewed 2+ pages. Subtracting the pairwise-overlap counts (S_2) corrects most of that overcounting but now slightly *undercounts* anyone who viewed 3+ pages (they got subtracted too many times). Truncating PIE at an odd number of correction-terms always overshoots; at an even number, always undershoots — that alternation is exactly the Bonferroni sandwich.

► *Speed trick / key lesson:* General rule: truncating the inclusion–exclusion sum after an *odd* number of terms (S_1, or $S_1 - S_2 + S_3$, ...) gives an upper bound; after an *even* number ($S_1 - S_2$, or $S_1 - S_2 + S_3 - S_4$, ...) gives a lower bound.

Problem 5 (When Boole's inequality becomes vacuous). A market-making algorithm runs 50 independent statistical tests daily, each with a 2% false-positive rate under the null. Bound $P(\text{at least 1 false positive})$ using Boole's inequality, then compute the exact value. Does Boole's bound tell you anything useful here?

Solution. Boole: $P(\bigcup A_i) \leq \sum P(A_i) = 50(0.02) = 1.0$. Exact (independent tests): $P = 1 - (0.98)^{50} \approx 1 - 0.3642 = \boxed{0.6358}$.
Boole's bound of "≤ 1.0" is *always* true for any probability and carries zero information here — it's vacuous.

★ *Intuition:* Boole's inequality is like saying "the total spent across 50 store visits is at most \$50 if each visit costs at most \$1" — true, but if you already knew total spending can't exceed \$50 anyway (because that's your whole budget), the statement tells you nothing new. The bound is only *useful* when $n \times p \ll 1$; once $n \times p$ approaches or exceeds 1, you must fall back on the exact complement formula.

► *Speed trick / key lesson:* Before quoting a union bound as your final answer, sanity-check $n \times p$ against 1. If it's not comfortably below 1, Boole's bound is decorative, not diagnostic — this exact failure mode is why serious multiple-testing corrections (Bonferroni-adjusted significance levels) exist.

Problem 6 (Boole's inequality as a genuinely tight approx-

imation). Among $n = 10$ people (365 equally likely birthdays), bound P(at least one shared birthday) via Boole's inequality and compare to the known exact value ≈ 0.1169.

Solution. There are $\binom{10}{2} = 45$ pairs, each sharing a birthday with probability $1/365$.

$$\text{Boole:} \quad P \le \frac{45}{365} \approx \boxed{0.1233}.$$

Exact value (from the birthday-problem formula) ≈ 0.1169. The Boole bound overshoots by only ≈ 0.0064 ($\approx 5.5\%$ relative error) — excellent here, in sharp contrast to Problem 5.

★ *Intuition:* Now compare to Problem 5's failure: here each of the 45 "pair-match" events is individually tiny ($1/365 \approx 0.27\%$) and there are far fewer of them relative to their size, so double-counted overlaps (three-way birthday coincidences) are rare enough to barely matter. Boole's inequality is reliable exactly when collisions between the events themselves are rare — the same logic that made Problem 5's bound vacuous (many, not-so-rare events) makes this one nearly exact (few, genuinely rare events).

▶ *Speed trick / key lesson:* Pairing Problems 5 and 6 side by side is the fastest way to internalize *when* to trust a union bound at Stage-1 speed: check whether the individual event probabilities are "birthday-small" or "false-positive-large" relative to how many you're summing.

I.C — Pairwise vs. Mutual Independence

Problem 7 (The classic pairwise-but-not-mutual counterexample). Two fair coins are flipped. Let $A = \{\text{coin 1 is heads}\}$, $B = \{\text{coin 2 is heads}\}$, $C = \{\text{the two coins match}\}$. Show A, B, C are pairwise independent but *not* mutually independent.

Solution. $\Omega = \{HH, HT, TH, TT\}$, each with probability $1/4$. $A = \{HH, HT\}$, $B = \{HH, TH\}$, $C = \{HH, TT\}$, each with probability $1/2$. $A \cap B = A \cap C = B \cap C = \{HH\}$, each with probability $1/4 = \frac{1}{2} \times \frac{1}{2}$. So all three pairwise products check out: **pairwise independent.** But $A \cap B \cap C = \{HH\}$, probability $1/4$, while $P(A)P(B)P(C) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$. Since $\frac{1}{4} \neq \frac{1}{8}$, A, B, C **are not mutually independent.**

★ *Intuition:* If you know A and B both happened (both coins are heads), you know C happened *for certain* (they obviously match) — full determinism, the opposite of independence. Pairwise checks are blind to this because they only ever look at two events at a time; the three-way interaction is invisible until you actually test the triple intersection.

▶ *Speed trick / key lesson:* "I checked every pair and they were all independent" is *not* a proof of mutual independence for $n \ge 3$ events — you must additionally verify the full $2^n - n - 1$ higher-order intersection products (or at minimum the top-level n-way one).

Problem 8 (Why this matters: a risk-model failure). A junior quant builds $X, Y \overset{\text{iid}}{\sim} \text{Bernoulli}(1/2)$ and $Z = X \oplus Y$ (XOR: $Z = 1$ iff $X \neq Y$). He verifies $P(X \cap Y) = P(X)P(Y)$, $P(X \cap Z) = P(X)P(Z)$, $P(Y \cap Z) = P(Y)P(Z)$ all hold, and concludes "all three are independent, so I can multiply their probabilities in my risk model." Show this conclusion is dangerously wrong.

Solution. Pairwise checks (all pass): $P(X{=}1, Y{=}1) = \frac{1}{4} = P(X{=}1)P(Y{=}1)$. $P(Z{=}1) = P(X \neq Y) = \frac{1}{2}$; $P(X{=}1, Z{=}1) = P(X{=}1, Y{=}0) = \frac{1}{4} = P(X{=}1)P(Z{=}1)$. By symmetry Y, Z also pass. Now the triple: $P(X{=}1, Y{=}1, Z{=}1)$ requires $X \neq Y$ *and* $X = Y = 1$ simultaneously — impossible. So

$$P(X{=}1, Y{=}1, Z{=}1) = 0,$$

$$\text{but} \quad P(X{=}1)P(Y{=}1)P(Z{=}1) = \tfrac{1}{8} \neq 0.$$

The gap here (0 vs. $\frac{1}{8}$) is even starker than Problem 7's — Z is in fact a *deterministic function* of (X, Y), about as dependent as a triple can be, yet it sails through every pairwise check.

★ *Intuition:* This is the quant version of "my three risk factors look uncorrelated pairwise, so I'll assume their joint tail risk multiplies out" — exactly the kind of reasoning that underpriced correlated mortgage-default risk in 2008. Pairwise independence tells you nothing about what happens when *all* the factors move together, which is precisely the scenario that blows up a portfolio.

▶ *Speed trick / key lesson:* If one variable can be reconstructed exactly from the others ($Z = X \oplus Y$), mutual independence is impossible *by construction*, no matter how clean the pairwise numbers look — always ask "can I derive any one of these from the others?" before trusting a pairwise-independence check.

II. Bayesian Updating & Conditioning

II.A — Conditional Probability

Problem 9 (Reading conditional probability off a 2×2 table). Of 500 trading days, 300 were high-volatility (V), 220 were positive-P&L (P), and 150 were both. Find $P(P \mid V)$, $P(V \mid P)$, and determine whether V, P are independent.

Solution. $P(V) = \frac{300}{500} = 0.6$, $P(P) = \frac{220}{500} = 0.44$, $P(V \cap P) = \frac{150}{500} = 0.3$.

$$P(P \mid V) = \frac{0.3}{0.6} = \boxed{0.5}, \qquad P(V \mid P) = \frac{0.3}{0.44} \approx \boxed{0.682}.$$

Independence check: $P(V)P(P) = 0.6 \times 0.44 = 0.264 \neq 0.3 = P(V \cap P)$, so **not independent** — volatility and positive P&L days co-occur *more* than chance predicts (positive association).

★ *Intuition:* $P(P \mid V)$ answers "restricting my attention only to high-vol days, what fraction were profitable?" — you've shrunk the sample space from all 500 days down to just the 300 high-vol days, then asked what fraction of *that* smaller space is also in P.

Problem 10 (Conditional probability is not symmetric). A fair die is rolled. $A = \{\text{even}\} = \{2, 4, 6\}$, $B = \{\text{at most } 2\} = \{1, 2\}$. Compute $P(A \mid B)$ and $P(B \mid A)$ and confirm they differ.

Solution. $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $A \cap B = \{2\}$, $P(A \cap B) = \frac{1}{6}$.

$$P(A \mid B) = \frac{1/6}{1/3} = \boxed{\tfrac{1}{2}}, \qquad P(B \mid A) = \frac{1/6}{1/2} = \boxed{\tfrac{1}{3}}.$$

Indeed $P(A \mid B) \neq P(B \mid A)$.

★ *Intuition:* This asymmetry is the entire reason Bayes' theorem needs to exist. "The probability it's raining given the streets are wet" and "the probability the streets are wet given it's raining" are obviously different questions (streets can be wet from a sprinkler) — confusing the two is called "confusion of the inverse," and it's the single most common Bayesian trap. Every problem in Section III below is a variation on this exact confusion.

▶ *Speed trick / key lesson:* If a problem seems to conflate $P(A \mid B)$ with $P(B \mid A)$, that's your cue: you need Bayes' theorem (Problem 12), not the plain conditional-probability definition.

II.B — Law of Total Probability

Problem 11 (LTP over a partition of order-flow sources). Desk A contributes 50% of order flow with a 2% toxic-fill rate; Desk B contributes 30% with a 5% rate; Desk C contributes 20% with a 10% rate. What fraction of all fills are toxic?

Solution. By the Law of Total Probability, partitioning on which desk the fill came from:

$$P(\text{Toxic}) = (0.5)(0.02) + (0.3)(0.05) + (0.2)(0.10)$$

$$= 0.01 + 0.015 + 0.02 = \boxed{0.045}.$$

★ *Intuition:* This is just a weighted average: each desk's toxicity rate gets weighted by how much volume it actually contributes. It's identical in spirit to computing a portfolio's overall return as the volume-weighted average of each asset's return.

Problem 12 (LTP with a hidden/random parameter — a genuine trap). A coin's type is random: w.p. 0.5 it's fair (p=0.5), w.p. 0.3 it's head-biased (p=0.7), w.p. 0.2 it's tail-biased (p=0.3). *The same coin* is flipped twice. Find $P(\text{both heads})$, and show why squaring the marginal $P(\text{heads})$ gives the *wrong* answer.

Solution. Correct method (LTP on the joint event, conditioning on coin type):

$$P(HH) = \sum_{\text{type}} P(\text{type})\, P(HH \mid \text{type})$$

$$= (0.5)(0.5)^2 + (0.3)(0.7)^2 + (0.2)(0.3)^2$$

$$= 0.125 + 0.147 + 0.018 = \boxed{0.290}.$$

Tempting but wrong method: first find the marginal $P(H) = (0.5)(0.5) + (0.3)(0.7) + (0.2)(0.3) = 0.52$, then square it: $0.52^2 = 0.2704 \neq 0.290$.

★ *Intuition:* The two flips are *not* unconditionally independent, even though they're independent *given* the coin type — because not knowing the coin's type means the first flip's outcome is itself evidence about which coin you're holding, which then shifts your expectation for the second flip. This is exactly how "regime uncertainty" induces correlation between asset returns even when each regime's returns are i.i.d. — a hidden common factor (here, coin identity; in markets, the macro regime) glues otherwise-independent-looking observations together.

▶ *Speed trick / key lesson:* Whenever a random experiment is repeated using the *same* hidden parameter across trials, the trials are correlated unconditionally — never just square/multiply the marginal probability. Always condition on the hidden variable first (LTP), *then* combine.

II.C — Bayes' Theorem (Standard & Odds Form)

Problem 13 (Standard Bayes' theorem — the base-rate surprise). A monitor flags "anomalous trading" with 95% sensitivity and a 2% false-positive rate. True anomalies occur in 1% of sessions. Given a session is flagged, what's the probability it's truly anomalous?

Solution.

$$P(\text{Flag}) = \underbrace{(0.95)(0.01)}_{\text{true pos.}} + \underbrace{(0.02)(0.99)}_{\text{false pos.}} = 0.0095 + 0.0198 = 0.0293.$$

$$P(\text{Anomaly} \mid \text{Flag}) = \frac{(0.95)(0.01)}{0.0293} = \frac{0.0095}{0.0293} \approx \boxed{0.324}\ (32.4\%).$$

★ *Intuition:* Even a "95%-accurate" test is mostly wrong when it fires, because true anomalies are so rare that the much larger pool of *normal* sessions still throws off enough 2%-rate false alarms to outnumber the real ones almost 2-to-1. It's the classic "haystack problem": a great needle-detector still finds mostly hay if the haystack is enormous relative to the needle.

Problem 14 (Odds-form Bayes — same problem, faster machinery). Re-solve Problem 13 using the *odds form*: Posterior Odds = Prior Odds × Bayes Factor.

Solution. Prior odds (Anomaly : Normal) $= \frac{0.01}{0.99} = \frac{1}{99}$. Bayes factor $= \frac{P(\text{Flag} \mid \text{Anomaly})}{P(\text{Flag} \mid \text{Normal})} = \frac{0.95}{0.02} = 47.5$. Posterior odds $= \frac{1}{99} \times 47.5 = \frac{47.5}{99} \approx 0.4798$. Convert back: $P = \frac{\text{odds}}{1+\text{odds}} = \frac{0.4798}{1.4798} \approx \boxed{0.324}$ — matches Problem 13 exactly.

★ *Intuition:* Odds form is a detective's mental math: "my prior suspicion was 1-in-99 against; this new clue is 47.5 times more likely if guilty than if innocent, so multiply the odds by 47.5." No denominators, no $P(\text{Flag})$ normalizing constant to compute — just multiply.

▶ *Speed trick / key lesson:* The odds form is strictly faster whenever you'll need to *chain* multiple pieces of evidence (see Problem 15) — the standard form forces you to recompute a fresh normalizing denominator $P(\text{evidence})$ every single time, while odds form just keeps multiplying Bayes factors.

Problem 15 (Sequential Bayesian updating — chaining odds). A second, independent monitor (90% sensitivity, 3% false-positive rate) *also* flags the same session as Problem 13/14. Given both flags (conditionally independent given the true state), find the posterior probability of a true anomaly.

Solution. Start from Problem 14's prior odds $\frac{1}{99}$. After flag 1: multiply by Bayes factor $\frac{0.95}{0.02} = 47.5 \to$ odds $= \frac{47.5}{99}$. After flag 2: multiply by Bayes factor $\frac{0.90}{0.03} = 30 \to$ odds $= \frac{47.5}{99} \times 30 = \frac{1425}{99} \approx 14.39$. Convert: $P = \frac{14.39}{1+14.39} \approx \boxed{0.935}$ (93.5%).

★ *Intuition:* This is precisely how a detective's confidence compounds across multiple independent clues: each new piece of matching evidence multiplies your odds by that clue's own Bayes factor, and confidence snowballs from "probably not" (1% prior) to "almost certainly" (93.5%) after just two moderately strong, independent signals agree.

▶ *Speed trick / key lesson:* This sequential-multiplication structure is exactly why real fraud/anomaly-detection systems stack multiple independent weak signals rather than relying on one strong one — and it's the fastest way to solve any "given multiple pieces of independent evidence" Bayes problem under time pressure.

III. Classical Bayesian Traps

Problem 16 (Monty Hall — rigorous Bayes derivation, then N-door generalization). 3 doors, 1 car, host knows the car's location and always opens a different, losing door, then offers a swap. Find $P(\text{win} \mid \text{switch})$ via Bayes' theorem (not handwaving), then generalize to N doors (host opens $N-2$ losing doors).

Solution. WLOG you pick door 1; host opens door 3. Prior: $P(\text{car} = i) = \frac{1}{3}$ each.

$$P(\text{opens } 3 \mid \text{car} = 1) = \tfrac{1}{2}, \quad P(\text{opens } 3 \mid \text{car} = 2) = 1,$$

$$P(\text{opens } 3 \mid \text{car} = 3) = 0.$$

$$P(\text{opens } 3) = \tfrac{1}{2} \cdot \tfrac{1}{3} + 1 \cdot \tfrac{1}{3} + 0 \cdot \tfrac{1}{3} = \tfrac{1}{2}.$$

$$P(\text{car} = 1 \mid \text{opens } 3) = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2}} = \tfrac{1}{3}\ (\text{stay}),$$

$$P(\text{car} = 2 \mid \text{opens } 3) = \frac{1 \cdot \frac{1}{3}}{\frac{1}{2}} = \boxed{\tfrac{2}{3}}\ (\text{switch}).$$

N-door generalization: $P(\text{win} \mid \text{stay}) = \frac{1}{N}$ always; a knowledgeable host eliminates $N-2$ losers, concentrating the remaining $\frac{N-1}{N}$ probability mass onto the single surviving unopened door:

$$P(\text{win} \mid \text{switch}) = \boxed{\frac{N-1}{N}}. \quad (\text{Check } N{=}3 : \tfrac{2}{3}.\ \checkmark)$$

★ *Intuition:* Picture 100 doors instead of 3: you pick one (1% chance of the car), the host — who *knows* where the car is — opens 98 losing doors, leaving your door and exactly one other. Does it feel like your original 1% pick suddenly deserves 50%? No: the host's knowledgeable elimination has funneled the other doors' entire 99% probability mass onto that one surviving door. Switching wins 99% of the time.

Problem 17 (Monty Fall — the host doesn't know). Same 3 doors, but now the host opens one of your two non-picked doors

uniformly at random, not knowing where the car is — and it happens to reveal a goat. Given this, find $P(\text{car in your door})$ vs. $P(\text{car in the other unopened door})$, and explain why the answer changes from Monty Hall despite an *identical-looking* observation.

Solution. WLOG you picked door 1, host randomly opened door 3 and it showed a goat. Compute $P(\text{opens 3 AND reveals a goat} \mid \text{car} = i)$ for each i: Car= 1: doors 2,3 both goats; host picks door 3 w.p. $\frac{1}{2}$, automatically a goat: joint $= \frac{1}{2}$. Car= 2: host picks door 3 w.p. $\frac{1}{2}$; since car is in door 2, door 3 is a goat: joint $= \frac{1}{2}$. Car= 3: if host picks door 3 (w.p. $\frac{1}{2}$), that would reveal the *car*, contradicting "goat revealed" — so this branch contributes 0 to our conditioning event.

$$P(\text{observed}) = \tfrac{1}{3}\left(\tfrac{1}{2}\right) + \tfrac{1}{3}\left(\tfrac{1}{2}\right) + \tfrac{1}{3}(0) = \tfrac{1}{3}.$$

$$P(\text{car} = 1 \mid \text{observed}) = \frac{\frac{1}{3}\cdot\frac{1}{2}}{\frac{1}{3}} = \tfrac{1}{2},$$

$$P(\text{car} = 2 \mid \text{observed}) = \frac{\frac{1}{3}\cdot\frac{1}{2}}{\frac{1}{3}} = \tfrac{1}{2}.$$

So it's $\boxed{\text{50-50}}$ — switching gives no advantage at all, unlike classic Monty Hall's $\frac{2}{3}$.

★ *Intuition:* The revealed door *looks* identical in both games (a goat, behind a door you didn't pick) — but *how* that goat came to be revealed carries completely different information. A knowledgeable host who deliberately avoids the car is silently telling you "I steered away from danger," which is real evidence. A host who got lucky by chance is telling you nothing beyond "this particular door wasn't the car" — and you already knew at least one of the other two doors had to be a goat. This is the sharpest illustration in all of probability that *the mechanism generating your evidence matters as much as the evidence itself.*

▶ *Speed trick / key lesson:* Before applying Monty Hall's "always switch" reflex, check: does the host know where the car is, and does the host deliberately avoid it? If either answer is no, the classic $\frac{2}{3}$ result does not apply.

Problem 18 (Bertrand's Box Paradox). Box GG has 2 gold coins, box GS has 1 gold + 1 silver, box SS has 2 silver coins. Pick a box uniformly at random, draw one random coin from it, and it's gold. What's the probability the *other* coin in that box is also gold?

Solution. Prior $P(\text{box}) = \frac{1}{3}$ each. $P(\text{gold} \mid GG) = 1$, $P(\text{gold} \mid GS) = \frac{1}{2}$, $P(\text{gold} \mid SS) = 0$.

$$P(\text{gold}) = \tfrac{1}{3}(1) + \tfrac{1}{3}\left(\tfrac{1}{2}\right) + \tfrac{1}{3}(0) = \tfrac{1}{3} + \tfrac{1}{6} = \tfrac{1}{2}.$$

$$P(GG \mid \text{gold}) = \frac{\frac{1}{3}\cdot 1}{\frac{1}{2}} = \boxed{\tfrac{2}{3}} \neq \tfrac{1}{2}.$$

★ *Intuition:* The naive "it's either box GG or box GS, so 50-50" answer counts *boxes* instead of *coins* — but a gold coin was twice as likely to come from box GG (which always shows gold) as from box GS (which shows gold only half the time). This is structurally identical to Monty Hall: observing "gold" is exactly like the host opening a door — it's informative evidence that should shift your box-probabilities unevenly, not split them evenly.

Problem 19 (Boy-or-Girl paradox — three phrasings, three different answers). A family has 2 children, $P(\text{boy}) = P(\text{girl}) = \frac{1}{2}$ independently. Find $P(\text{both boys})$ given: (a) "at least one is a boy"; (b) "the elder is a boy"; (c) "you meet a randomly chosen one of the two children, and it's a boy."

Solution. Sample space $\{BB, BG, GB, GG\}$ (elder,younger), each $\frac{1}{4}$. **(a)** Conditioning on "at least one boy" removes GG: $P(BB \mid \text{at least 1 boy}) = \dfrac{1/4}{3/4} = \boxed{\frac{1}{3}}$. **(b)** Conditioning on "elder is a boy" restricts to $\{BB, BG\}$: $P(BB \mid \text{elder} = B) = \dfrac{1/4}{2/4} = \boxed{\frac{1}{2}}$. **(c)** By LTP over family type: $P(\text{meet boy} \mid BB) = 1$, $P(\text{meet boy} \mid BG) = P(\text{meet boy} \mid GB) = \frac{1}{2}$, $P(\text{meet boy} \mid GG) = 0$.

$$P(\text{meet boy}) = \tfrac{1}{4}(1) + \tfrac{1}{4}\left(\tfrac{1}{2}\right) + \tfrac{1}{4}\left(\tfrac{1}{2}\right) + \tfrac{1}{4}(0) = \tfrac{1}{2}.$$

$$P(BB \mid \text{meet boy}) = \frac{\frac{1}{4}(1)}{\frac{1}{2}} = \boxed{\tfrac{1}{2}}.$$

So (a) gives $\frac{1}{3}$, while (b) and (c) both give $\frac{1}{2}$ — (c) is secretly equivalent to (b), *not* to (a).

★ *Intuition:* This "paradox" dissolves once you notice the three statements describe genuinely different *sampling procedures*, not just different phrasings of the same fact. (a) is a fact about the unordered family (something is true of the pair, with no information on how you learned it or which child is which). (b)/(c) both single out one *specific* child (by birth order, or by literally meeting them) and tell you that particular child's sex — which halves the space cleanly in the same way. The lesson: before conditioning, always ask "what *exact* random experiment generated this piece of information?"

▶ *Speed trick / key lesson:* When two "boy-or-girl"-style phrasings feel equivalent but give different textbook answers, check whether one identifies a *specific* child (elder, met-at-random, first-through-the-door) versus merely asserting existence over the unordered pair — that distinction alone flips $\frac{1}{2} \leftrightarrow \frac{1}{3}$.

Problem 20 (Base-rate neglect via natural frequencies — a debiasing technique). A fraud test has 99% sensitivity and 99% specificity; true fraud occurs in 1-in-10,000 trades. Using *natural frequencies* (whole numbers, not probabilities) over 1,000,000 trades, find what fraction of *flagged* trades are truly fraudulent.

Solution. Truly fraudulent: $1{,}000{,}000 \times \frac{1}{10{,}000} = 100$. Truly clean: 999,900. True positives (flagged & fraudulent): $100 \times 0.99 = 99$. False positives (flagged & clean): $999{,}900 \times 0.01 = 9{,}999$. Total flagged: $99 + 9{,}999 = 10{,}098$.

$$P(\text{truly fraudulent} \mid \text{flagged}) = \frac{99}{10{,}098} \approx \boxed{0.0098} \quad (\approx 0.98\%).$$

★ *Intuition:* Despite "99% accurate" sounding airtight, fewer than 1-in-100 flags are real fraud — because among a million trades there are simply so many more clean trades (999,900) than fraudulent ones (100) that even a tiny 1% false-positive rate generates ten times more false alarms than there are true frauds to find. Counting actual people/trades out of a round number (Gigerenzer's "natural frequency" method) is exactly how doctors, judges, and analysts are trained to avoid this trap — it sidesteps the fraction arithmetic that hides the base rate, and you can literally count on your fingers instead of manipulating decimals.

▶ *Speed trick / key lesson:* At Stage-1 speed, if a "how many false alarms" question feels fraction-heavy, translate to natural frequencies over a convenient round number (1,000 or 1,000,000) first — it's faster *and* far harder to sign-flip or misplace a term in than the raw Bayes formula.

Drill order for review: re-derive Problems 12, 16, 17, and 19 from scratch (closed-book). Together they carry the four recurring traps of this whole chapter — hidden-parameter correlation, knowledgeable vs. random evidence generation, and sampling-procedure ambiguity — that reappear constantly in Stage-1 "obvious answer is wrong" questions.